

mbariml: a curation pipeline for turning deep-sea imagery and video into object-detection training data

Lonny Lundsten

Monterey Bay Aquarium Research Institute

lonny@mbari.org

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Abstract

Training an object detector for deep-sea ecology is rarely limited by modelling. It is limited by annotation. A single ROV dive produces hours of footage in which the animals of interest are sparse, faint, and hard to identify, and every improvement to the detector requires another round of labelling. *mbariml* is a pipeline built around that loop rather than around the model. It takes any Ultralytics YOLO model, runs it over still images or video, stores every detection as a reviewable region of interest, groups those regions by visual similarity so that a human can accept or reject them in bulk, and exports the result as training data. The human review stage is the centre of the design: an annotator can validate, relabel, resize, delete, and draw entirely new localizations, and every one of those edits is written back to the same database the detector wrote to. Video receives particular attention, since an animal held in frame for three hundred frames should become one training example rather than three hundred. We describe the pipeline stage by stage, and examine the rule we use to pick which frame of a track becomes the training crop. That rule rests on an annotator's judgement that detections are correct more often mid-track. We find that detector confidence does not follow the same pattern, which suggests confidence is a poor stand-in for correctness on this footage, and argues for choosing frames by position rather than by score.

1 The problem

Consider a typical benthic survey. An ROV flies a transect and records video, or a camera platform collects stills on a timer. Somewhere in that footage are the animals an ecologist cares about: sponges, corals, anemones, holothurians, the occasional fish. They are sparse. They are frequently low-contrast against sediment. Identifying them to a useful taxonomic level is expert work, and there is far more footage than expert time.

A detector helps, but not as much as one might hope, and it introduces problems of its own. Run it at a threshold low enough to catch faint animals and it returns a large number of false positives that someone must reject. Run it conservatively and it silently misses the things that most need finding. Either way the output is a pile of boxes that has to be checked, and checking them one at a time is no faster than annotating from scratch.

What actually helps is a loop. Over-detect deliberately, group the results so that hundreds of similar detections can be judged together, let a human correct what the detector got wrong, and feed the corrected set back into training. The work in building such a loop is mostly not in the model. It is in keeping the data in one place, making the human's time count, and getting the result back out in a form a trainer will accept.

2 What the pipeline does

mbariml is a command-line tool with a graphical review stage, organised into four phases (Figure 1). The sequence is:

1. Point any Ultralytics YOLO model at a folder of images or at video.
2. Get back one row per detection, each carrying its own cropped image.
3. Optionally embed and cluster those crops, then relabel a whole cluster at once using its dominant label.

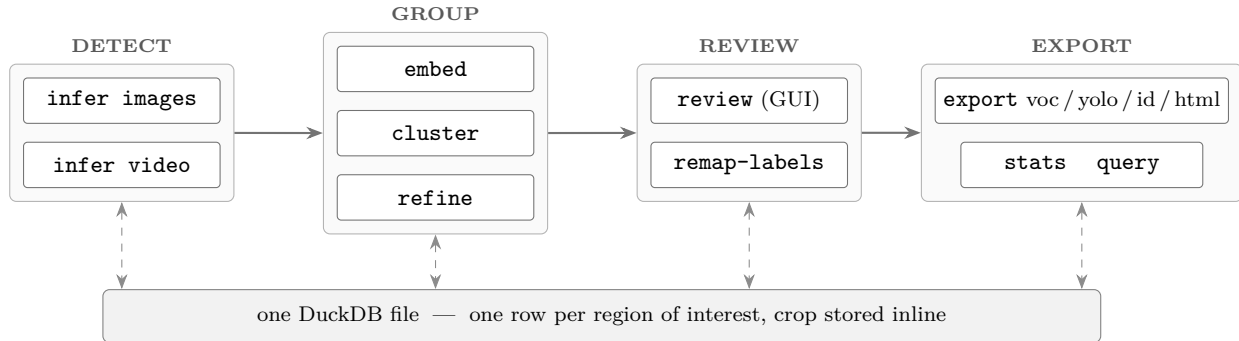


Figure 1: The four phases. All of them read and write the same table, so any stage’s output can be fed to any other stage that needs what it contains.

4. Review the result by hand: validate, relabel, adjust boxes, delete mistakes, and draw in the detections the model missed.
5. Export training data, summary statistics, or a browsable gallery.

Every one of those steps reads and writes the same DuckDB file, containing a single flat table in which one row is one region of interest. There is no per-stage intermediate format. A consequence worth stating plainly: no stage requires any particular earlier stage to have run, so pointing the video command at new footage and going straight to review is as valid as running the whole chain.

Each row stores its crop inline as a JPEG blob. Embedding, clustering, and the review mosaic therefore never touch the original imagery, which matters when the original is a multi-gigabyte video file or a network volume that is not currently mounted. The full frame is read only when a reviewer wants context or edits a box.

3 Detection: any Ultralytics model, stills or video

The detection stage is a thin layer over Ultralytics YOLO [1], which does all of the model loading, inference, and multi-object tracking. Any model Ultralytics can load will work, including the reader’s own fine-tuned weights; nothing in the pipeline is tied to a particular architecture or class list. Class names are read from the model itself.

Two presets encode the intent of a run. The `curate` preset uses a confidence threshold of 0.005 and a large inference size, and is meant for building training data: it over-detects on purpose, because a permissive threshold can be filtered afterwards while a strict one discards detections that cannot be recovered without a full re-run. The `predict` preset uses 0.08 and is meant for believable inference over new imagery. Both are starting points rather than fixed values: confidence, IoU, inference size, and the rest are ordinary options, and passing one overrides whatever the preset would have supplied.

For each detection the pipeline records the box, the class, the confidence, a blur score computed from the crop, and the crop itself.

4 Video: one region of interest per track

Video needs different treatment from stills. An animal that stays in frame for three hundred frames produces three hundred detections, and those are not three hundred training examples. They are one animal, photographed repeatedly. Left alone they overwhelm clustering, and they make review tedious in a way that causes annotators to stop paying attention.

The video command therefore runs Ultralytics’ tracker and keeps a single region of interest per track. On a ten-second clip this reduced 711 detections to 7 reviewable observations. The reduction is the point: what reaches the human is a list of animals, not a list of frames.

Because a track’s representative frame cannot be chosen until the track has finished, this runs as two passes. The first tracks the whole video and retains only per-track metadata, so memory stays proportional to the number of open tracks rather than to the length of the video. The second sweeps the file once and

extracts only the chosen frames. That sweep reads forward rather than seeking: frame-accurate seeking is unreliable on long-GOP encodings, where a seek lands on the nearest keyframe and would quietly pair a detection’s box with the wrong pixels. A forward sweep is exact, and we verified it by encoding frame numbers into the pixels of a test video and confirming that the frames returned carried the numbers requested.

Selected frames are written to disk as ordinary JPEGs, and the row points at the extracted frame. Video-derived rows are then indistinguishable from image-derived rows everywhere downstream, so review, embedding, clustering, and all four exporters work on video without containing any video-specific code. The link back to the footage is kept in separate columns, which is what lets the review GUI open the source video at the moment of the detection.

5 Grouping: embeddings and clusters

Clustering is optional but changes the economics of review. Each crop is embedded with DINOv3 (ViT-L/16), and the embeddings are clustered with EVoC. Each cluster is then labelled with the dominant class among its members, which propagates one decision across everything in the group. A reviewer looking at a page of visually similar sponges can accept them together instead of one at a time.

The same embeddings drive a similarity search in the GUI. Right-clicking any region of interest re-ranks the entire database by cosine similarity to it, which turns “show me the rest of these” into a single action. Embeddings are mean-centred on the ranking pool before normalisation; without that step the component common to nearly all deep-sea imagery dominates the dot product and compresses similarity into a narrow, weakly separated band.

The search reports the size of the pool it ranked, which is less pedantic than it sounds. Only regions that already carry an embedding can be ranked, so an embedding pass that was interrupted or deliberately limited silently reduces the search to whatever happened to be embedded. The ranked count and the matching count are displayed together—“all 8,412 matching regions” against “3,001 of 8,412”—because the two situations are otherwise indistinguishable from inside the interface, and the second is easily misread as the search having covered only the page in view.

Naming the clusters turns out to need care. The obvious approach, and the one we started with, is to label a cluster after the dominant original class among its members. That assumes the detector has enough classes for the label to distinguish one cluster from another, which is not always true. A single-class detector—MBARI’s Megalodon, for instance, which reports only `object`—makes every cluster’s dominant label the same string, so writing it back collapses the clustering that was just computed into one undifferentiated label. The structure survives only in the raw cluster number, and the review GUI, which filters and sorts on the curated label, can no longer separate the groups at all.

Clusters are therefore named with an index appended whenever one label wins more than one cluster: a single-class run yields `object_1`, `object_2`, and so on. On a 147-region run from a single-class model, six clusters that previously all became `object` now come back separately labelled. The same rule helps multi-class runs, where several clusters of the same taxon would otherwise be flattened together; on the same regions with the detector’s real labels, two singleton clusters keep their bare names while four distinct sponge clusters become `Hexactinellida_1` through `_4`. Merging them afterwards is one bulk relabelling command, whereas recovering a distinction that was thrown away is not.

Which label the naming vote reads is itself a choice, and defaulting it to the detector’s raw class is wrong as soon as a database has been reviewed. The human decisions live in the curated label, and on a single-class model the raw class is the same useless string for every region, so re-clustering a reviewed database discarded the review twice over: the write-back replaces the curated label, and the replacement was voted on from a column the reviewer never touched. The naming vote can now be pointed at the curated label instead, falling back to the raw class for regions not yet reviewed. On a 300-region database curated into three taxa, the default returns three indexed `object` names while the curated source returns the three taxon names.

This is worth stating alongside a caveat the command cannot enforce: the write-back replaces the curated label for every embedded region, including ones already marked verified, and does not clear the verified flag—so those rows afterwards appear reviewed while carrying a machine-generated name. Clustering experiments therefore belong on a copy of the database. The copy carries the embeddings, so the expensive stage is not repeated and each iteration costs seconds.

When the first clustering pass lumps things too coarsely, a refine step re-clusters a single label into finer

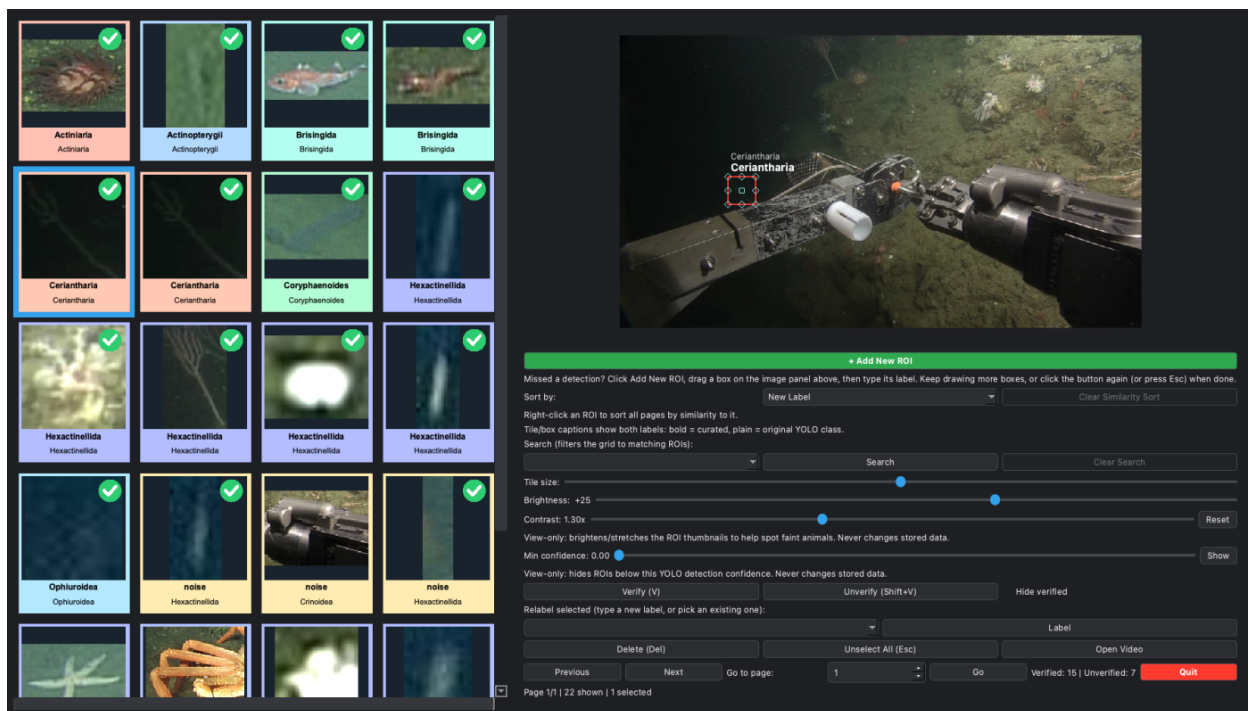


Figure 2: The review GUI. Left: the mosaic of regions of interest, tinted by label, with a badge on those already checked. Right: the full source frame with every detection on it drawn as an editable box, above the controls. Brightness and contrast sliders stretch the thumbnails without altering stored data, which matters for faint animals against sediment. “Open Video” is active here because this region came from video, and opens the footage at the moment of the detection.

groups, which is the usual remedy for a large catch-all bucket.

6 Review: the human in the loop

Everything above exists to make this stage efficient. The detector proposes; the annotator decides. The review GUI (Figure 2) presents regions of interest as a mosaic of crops, tinted by label, with a badge marking those that have been checked. An annotator working through a page can:

- **Validate** a detection, marking it reviewed. Applying a label counts as review, since labelling something is itself an act of checking it.
- **Relabel** one region or a selection of hundreds, either by typing a new concept or picking one already in use in this database.
- **Adjust** a box by dragging its handles on the full-resolution frame. The stored crop is regenerated to match, immediately.
- **Delete** false positives, individually or in bulk.
- **Add** a localization the model missed entirely, by drawing a box on the frame and naming it. Its embedding is computed in the background using the same model and preprocessing as the batch path, so a hand-drawn region is directly comparable to every other one and participates in similarity search straight away.

Two details turned out to matter more than expected. The first is that the right-hand panel draws *every* detection on the source frame, not only the ones on the current page, because detections from the same frame are often scattered across different pages by sorting. Without this, a reviewer editing a box has no idea what else was found on that image. The second is a pair of brightness and contrast sliders. Deep-sea imagery is

frequently dark and low-contrast, and animals that are invisible at native exposure become obvious when stretched. The adjustment applies to the displayed thumbnails only and never modifies stored data.

A separate bulk relabelling command handles taxonomy revisions across an entire database, applying all renames in a single pass so that a file which swaps two labels does what it says.

7 Choosing which frame represents a track

Reducing a track to one region raises the obvious question of which frame to keep. Our default takes the highest-confidence detection from the middle third of the track.

The reasoning comes from reviewing this kind of footage. An animal enters the field of view at the edge of the frame, passes through the region where the camera is best focused and best positioned, and then leaves. Detections made during that middle interval are, in our annotators' experience, *correct* more often: the identification survives expert review more frequently than one made as the animal entered or left. Figure 3(a,b) shows a track that behaves this way, with the crops from mid-transit better framed and sharper than those at either end.

That claim is about correctness as judged by a person, which is not the same quantity as the detector's own confidence score. The difference is worth keeping in view, because the two can disagree: a model can be confidently wrong about a poorly framed animal, and unsure about a well-framed one it happens to have seen rarely in training.

We measured the confidence side across 19 tracks from three clips, and it does *not* show a middle-third peak. Confidence is highest in the first third for 9 of the 19 tracks, in the middle third for 4, and in the last third for 6, and the middle third has the lowest mean of the three (0.301, against 0.333 for the first third; Figure 3(c)). Some of this is geometry: an ROV often approaches an animal and stops, or passes close enough that the view keeps improving until the track ends, so the best frames are the last ones rather than the middle ones. Two of the longest tracks we examined behave that way.

The two observations are not in conflict, and the disagreement is the interesting part. If human-assessed correctness does peak mid-track while confidence does not, then confidence is not tracking correctness on this footage, and picking a frame by score alone would be the wrong thing to do. A positional rule is then doing work that a confidence threshold cannot.

We have not closed the loop on this. Establishing the correctness claim properly means reviewing detections sampled from the first, middle, and last thirds of many tracks and recording how often each is right, which is expert time we have not yet spent. Until then the middle-third rule stands as an informed default rather than a measured one, and the alternatives—highest confidence anywhere in the track, the sharpest frame in the middle third, or the exact centre frame—are selectable per run.

8 Getting the data back out

The export stage writes Pascal VOC XML, YOLO-format label files with the accompanying names file, *.id sidecar files placed beside the source images for MBARI's own tooling, and a paginated HTML gallery for quick visual checks. The VOC and YOLO exports also emit a manifest of the source images and a small standalone script that copies them, since annotation files are of no use to a trainer without the imagery beside them.

The YOLO export additionally writes the train, validation and test image lists, which completes the dataset rather than leaving the last step to be done by hand. Two decisions in how they are written are worth recording. The listed paths are relative to the dataset directory, so the directory refers only to itself and can be archived or moved to a training machine without any path being rewritten—absolute paths recorded on the exporting machine could not survive that move. And the split is taken over images, never over individual detections. Two crops of the same frame placed on opposite sides of the train/validation boundary share their background, their illumination and often the same individual animal, which inflates validation scores on transect imagery where consecutive frames already overlap heavily. The split is seeded by default for the same reason the rest of the pipeline is reproducible: a re-export after correcting a handful of labels should not quietly reshuffle which images were held out, since that would make models trained either side of the re-export incomparable.

Two commands support analysis rather than training. A statistics command reports label counts and per-image detection counts, and writes a label-by-image matrix suitable for ecological work such as per-image

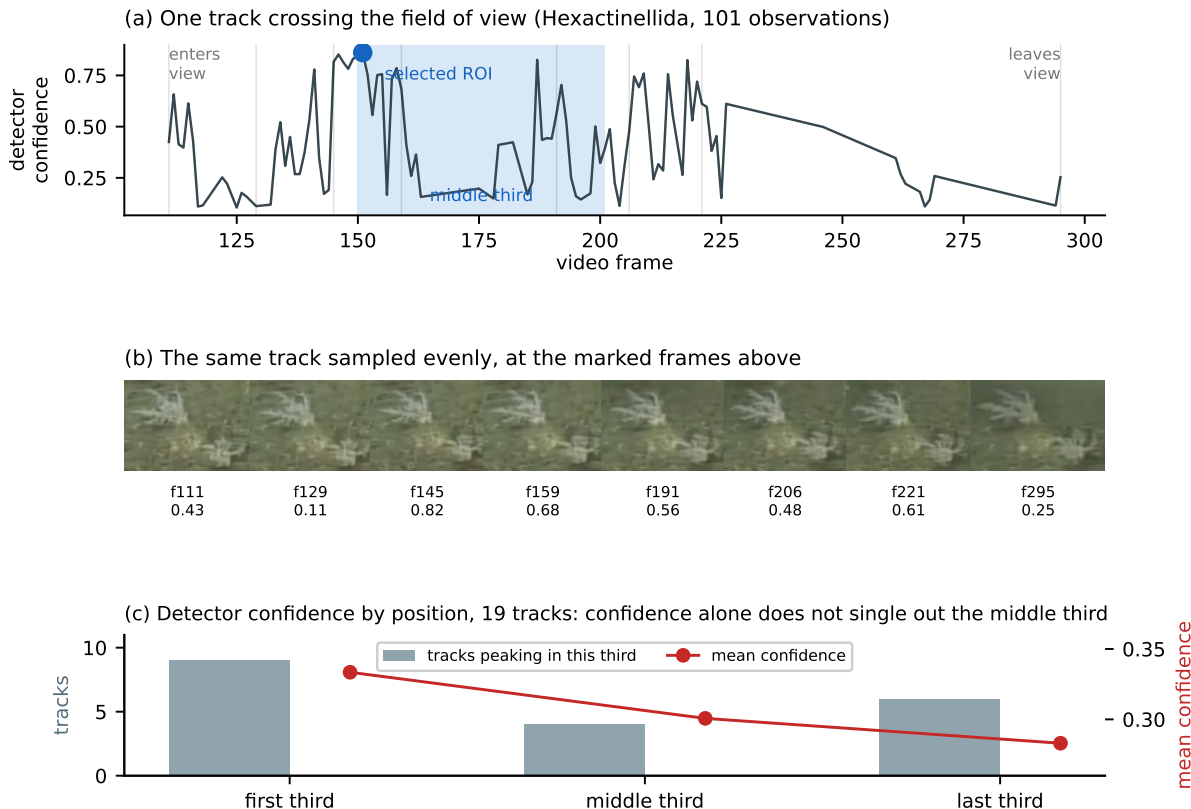


Figure 3: Choosing a frame from a track. (a) Detector confidence across one 101-observation track, with the middle third shaded and the selected region marked. (b) The same track sampled evenly at the frames marked in (a); the animal is best framed part-way through the transit. (c) Detector confidence across 19 tracks from three clips does not single out the middle third: it peaks in the first third more often than in the middle, and the middle third has the lowest mean. Confidence is a different quantity from the human-assessed correctness the rule is based on.

richness or frequency of occurrence. An ad hoc query command runs arbitrary SQL against the database, which is often the fastest way to answer a question the tool does not anticipate.

9 Limitations

The evaluation here is internal and small. We report data-reduction and throughput figures, not detection accuracy. The track-selection rule in particular rests on an annotator’s judgement that has not been quantified: the experiment that would settle it is a review pass over detections sampled from each third of many tracks, scoring how often each is correct, followed by training runs on datasets built under each rule. The longest video processed end to end is short; the first pass retains only metadata and should scale flatly with duration, but this is unverified at full dive length. Clustering quality depends on EVoC parameters that in practice need tuning per deployment. Because DuckDB takes an exclusive lock on the database file, two people cannot curate the same database at once, and combining separately curated databases is a manual merge.

10 Availability and credits

`mbariml` is written in Python and available at <https://github.com/mbari-org/mbari-ml>.

Detection and tracking are provided entirely by **Ultralytics YOLO** [1], which the pipeline wraps rather than reimplements; model loading, inference, and the multi-object trackers are all theirs. The review GUI’s mosaic rendering, threading, and box-overlay machinery are adapted from MBARI’s *vars-gridview* [2] under the MIT licence. Embeddings use DINOv3 [3] via `timm`, clustering uses EVoC [4], and storage is DuckDB [5].

The database layer, the video ingest and track-reduction stages, the export formats, and the additions to the review GUI described here are original to this work.

References

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